Leonardo Franchi

PhD Student in Mathematics
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Curriculum Vitae

Education

Oct. 2025–present **PhD in Mathematics**, University of Cambridge, Cambridge, UK.
Supervisor: Timothy Gowers.

Oct. 2023–Jul. 2025 **Master's Degree in Mathematics**, University of Padua, Padua, Italy.
Final grade: 110/110 cum laude. Thesis: *Osgood meets DiPerna–Lions*. Advisor: Guido De Philippis.

Oct. 2020–Oct. 2025 **Galilean Diploma**, Galilean School of Higher Education, University of Padua, Padua, Italy.

Oct. 2020–Feb. 2023 **Bachelor's Degree in Mathematics**, University of Padua, Padua, Italy.
Final grade: 110/110 cum laude.

Submitted papers and preprints

1. Guido De Philippis and Leonardo Franchi, *Osgood meets Ambrosio–DiPerna–Lions*, arXiv:2607.28118 (2026).
2. Valentino Badalucco and Leonardo Franchi, *Fractal uncertainty principle over* $\mathbb{Q}_p$, arXiv:2607.11534 (2026).
3. Leonardo Franchi, W. T. Gowers and Fredy Yip, *Product-free subsets of* $(0, 1)$, arXiv:2607.06073 (2026).
4. Gaia Carenini and Leonardo Franchi, *A unified abstract regularity lemma*, arXiv:2606.06192 (2026).
5. Gaia Carenini and Leonardo Franchi, *A strengthening of Chang's lemma*, arXiv:2605.07916 (2026).

Invited Talks

26 Oct. 2026 *Product-free subsets of* $(0, 1)$ *and a strengthened Brunn–Minkowski inequality*.
The Swinnerton-Dyer Seminar, University of Warwick, Coventry, UK.

12 Oct. 2026 *Product-free subsets of* $(0, 1)$ *and a strengthened Brunn–Minkowski inequality*.
Junior Number Theory Seminar, University of Oxford, Oxford, UK.

29 Sep. 2026 *Product-free subsets of* $(0, 1)$ *and a strengthened Brunn–Minkowski inequality*.
Combinatorics Seminar, University of Bristol, Bristol, UK.

26–28 Jun. 2026 *A strengthening of Chang's lemma*.
WOMBL 18 (Warwick–Oxbridge–Manchester–Bristol–Lancaster Meeting on Additive Combinatorics and Analytic Number Theory), Ambleside Campus, University of Cumbria, UK.

Awards and Funding

2025-2029 Trinity Cambridge Research Studentship, supported by the Isaac Newton Trust.

2024 First Prize, International Mathematics Competition for University Students, Blagoevgrad, Bulgaria.

2020 First place, Italian Mathematical Olympiad.

Teaching and Academic Experiences

Michaelmas 2026 **Undergraduate Supervisor, Optimisation**, Queens' College, University of Cambridge.

Lent 2026 **Undergraduate Supervisor, Probability (Part IA)**, Queens' College, University of Cambridge.

Dec. 2025 **Examiner for Undergraduate Mathematics Admissions**, Queens' College, University of Cambridge.

Jun. 2023, 2024 **Lecturer, International Mathematics Summer Camp**, Beijing Institute of Mathematical Sciences and Applications (BIMSA), Beijing, China.